CUHKFAA Chan Chun Ha Secondary School

First Term Examination (2025-2026)

Secondary Two MATHEMATICS

PAPER 2

10:05 a.m. - 10:50 a.m.

12-12-2025

Time allowed: 45 minutes

Name:_____(___) Class:_____ Marks:_____/**30**

1. Read the instructions carefully on the Question Paper.

2. When told to read this paper, you should check that the question paper has no missing pages. Look for the words '**END OF PAPER'** on the last page.

3. All questions carry equal marks.

4. **ANSWER ALL QUESTIONS.** You are advised to mark all the answers on the Multiple-choice Answer Sheet with an HB pencil, so that wrong marks can be completely erased with a clean rubber.

5. You should mark only **ONE** answer to each question. Two or more answers will score **NO MARKS**.

6. No marks will be deducted for wrong answers.

There are 20 questions in Section A and 10 questions in Section B.

The diagrams in this paper are not necessarily drawn to scale.

Choose the best answer for each question.

Section A (1 – 20)

1. Which of the following is an identity?

   A. $2a = 2 + a$
   B. $3b - 1 = -(1 - 3b)$
   C. $2c + 3 = 3 - 2c$
   D. $5d - 8d = 3d$

2. If $2(x + P) - 5 \equiv Qx - 7$, find the constants P and Q.

   A. $P = -1, Q = 2$
   B. $P = 2, Q = -1$
   C. $P = 2, Q = 2$
   D. $P = 6, Q = 2$

3. Which of the following is/are the factor(s) of $16x^2 - 8xy + y^2$?

   I. 16
   II. $4x + y$
   III. $4x - y$

   A. I only
   B. II only
   C. III only
   D. I and III only

4. Which of the following is correct?

   A. $9x - 3y = 3(x - y)$
   B. $-9x - 3y = -3(3x - y)$
   C. $-9x + 3y = -3(3x + y)$
   D. $9x + 3y = 3(3x + y)$

5. Factorize $2h^3 + 4h^5$.

A. $2h^3(1 + h^2)$
B. $2h^3(1 + 2h^2)$
C. $2h^5(h^2 + 2)$
D. $2h^6(1 + 2h^2)$

6. Factorize $h - 8 + (h - 8)(8h + 1)$.

A. $(h + 8)(8h + 2)$
B. $(h - 8)(8h + 2)$
C. $(h + 8)(- 8h + 2)$
D. $(h - 8)(- 8h + 2)$

7. Factorize $100m^2 - 9n^2$.

A. $(10m - 3n)^2$
B. $(10m + 3n)(10m - 3n)$
C. $(100m + 9n)(100m - 9n)$
D. $10(m + 3n)(m - 3n)$

8. Which of the following has a as the subject of the formula?

A. $\frac{a}{3} = 1 - 2b$
B. $ab = 7$
C. $a^2 = b - a$
D. $a = 2b^2 + 3$

9. Simplify $\frac{a^2b}{a^4b^2}$.

A. $\frac{1}{2a^2b}$
B. $\frac{1}{2a^2b^2}$
C. $\frac{1}{a^2b}$
D. $\frac{1}{a^2b^2}$

10. Simplify $\frac{4c-3d}{3a+3b} \times \frac{a+b}{4c-3d}$.

A. $\frac{1}{3}$

B. $\frac{1}{3(a+b)}$

C. $\frac{2}{3(a+b)}$

D. $\frac{2(4c-3d)}{3(a+b)}$

11. Simplify $\frac{7}{a-b} + \frac{5}{a-b}$.

A. $\frac{12}{a-b}$

B. $\frac{2}{b-a}$

C. $\frac{12}{a+b}$

D. $\frac{2}{b-a}$

12. Given that $p = \frac{1}{2q} - \frac{1}{r}$, find the value of p if $q = 6$ and $r = 8$.

A. $\frac{1}{4}$

B. $\frac{1}{24}$

C. $-\frac{1}{4}$

D. $-\frac{1}{24}$

13. If $2x - 3y = 6z$, then $y =$

A. $\frac{2x - 2z}{3}$.

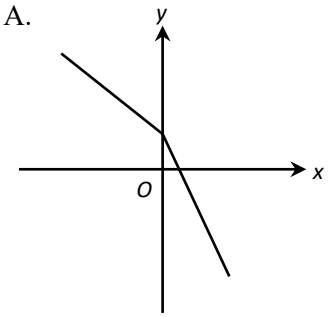
B. $\frac{2x - 6z}{3}$.

C. $\frac{6z - 2x}{3}$.

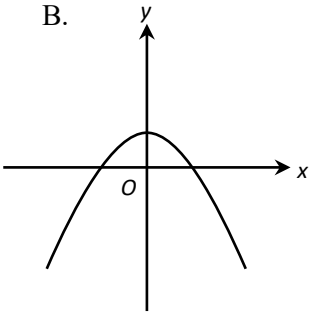
D. $\frac{2x + 6z}{3}$.

14. Which of the following may represent the graph of the equation $2x + 2y = 1$?

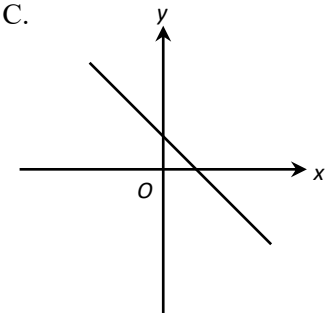
A.



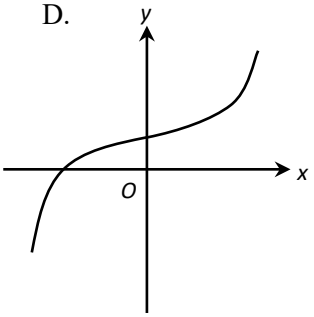
B.



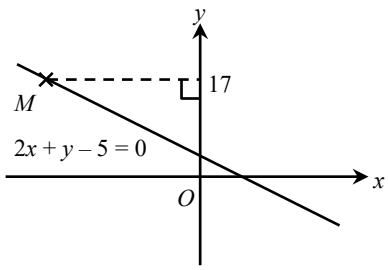
C.



D.



15. The figure shows the graph of the equation $2x + y - 5 = 0$.



M is a point lying on the graph. Find the x-coordinate of M.

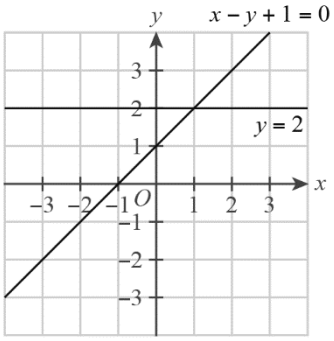
A. 0

B. –3

C. –5

D. –6

16. According to the figure, find the solution of the simultaneous equations $\begin{cases} x-y+1=0 \\ y=2 \end{cases}$.



A. $x=0,\ y=2$

B. $x=1,\ y=2$

C. $x=2,\ y=1$

D. $x=2,\ y=2$

17. If $\begin{cases} 2x-3y=16 \\ 5x+y=6 \end{cases}$, then $y=$

A. –5.

B. –4.

C. 2.

D. 16.

18. Solve the simultaneous equations $\begin{cases} 3x-7y=2 \\ 21y-9x=7 \end{cases}$.

A. $x=1, y=\frac{1}{7}$

B. $x=\frac{5}{3}, y=1$

C. No solutions

D. Infinitely many solutions

19. x and y are two numbers, where x is smaller than y. If the sum of x and y is 20 and their difference is 6, find the values of x.

A. 7

B. 8

C. 12

D. 13

20. Factorize $2(m-n)^2 + 2(n-m)$.

A. $(m-n)(m-n+1)$

B. $(m-n)(m-n-1)$

C. $2(m-n)(m-n+1)$

D. $2(m-n)(m-n-1)$

Section B (21 – 30)

21. Expand $\left(\frac{m}{3}-n\right)\left(n+\frac{m}{3}\right)$.

A. $\frac{m^2}{6}-n^2$

B. $\frac{m^2}{9}-n^2$

C. $n^2-\frac{m^2}{9}$

D. $n^2+\frac{m^2}{9}$

22. Simplify $(x+2y)^2 \times \frac{6y}{z(x+2y)} \div \frac{y(4y+2x)}{z^3}$.

A. $3z^2$

B. $3yz^2$

C. $\frac{z^2}{3}$

D. $\frac{yz^2}{3}$

23. If x equals half of y, and the sum of 4 times x and y is 48, then

A. $\begin{cases} x = \dfrac{y}{2} \\ 4x + y = 48 \end{cases}$

B. $\begin{cases} x = 2y \\ 4(x + y) = 48 \end{cases}$.

C. $\begin{cases} x = 2y \\ 4x + y = 48 \end{cases}$

D. $\begin{cases} x = \dfrac{y}{2} \\ 4(x + y) = 48 \end{cases}$.

24. If $a = \dfrac{2b - c}{4d}$, which of the following must be correct?

I. $b = \dfrac{4ad + c}{2}$

II. $c = 2b - 4ad$

III. $d = \dfrac{2b - c}{4a}$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

25. There are 300 seats available for the buffet in a hotel. The cost of the buffet for each adult is \$480, while the cost for each child is \$280. If the hotel had a full house for the buffet yesterday with a total income of \$136 000, how many children were there yesterday?

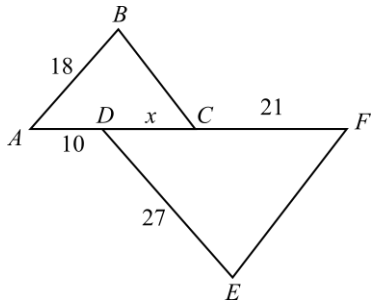
A. 25

B. 20

C. 35

D. 40

26. In the figure, $ADCF$ is a straight line. If $\triangle ABC \sim \triangle DEF$, find x.



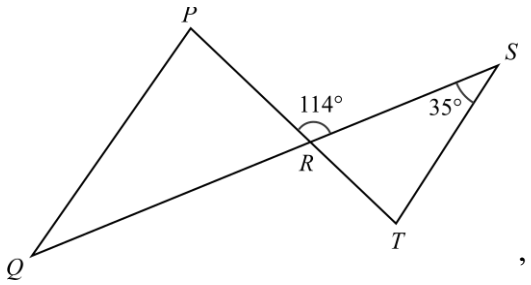
A. 4

B. 6

C. 12

D. 14

27. In the figure, PRT and QRS are straight lines, and $\triangle PQR \sim \triangle TSR$. Find $\angle QPR$.



,

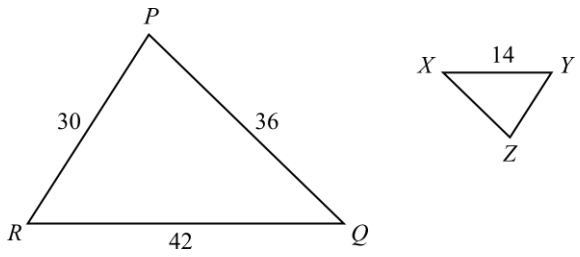
A. 35°

B. 66°

C. 79°

D. 89°

28. In the figure, $\triangle PQR \sim \triangle ZXY$. Find XZ and YZ.



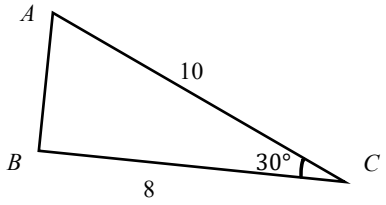
A. $XZ = 2$, $YZ = 8$

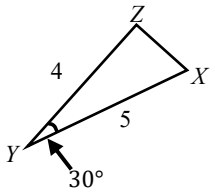
B. $XZ = 8$, $YZ = 2$

C. $XZ = 10$, $YZ = 12$

D. $XZ = 12$, $YZ = 10$

29. Refer to the figure. Which of the following is correct?





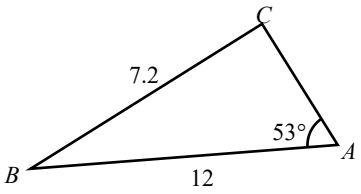
A. $\triangle ABC \sim \triangle XZY$ *(AAA)*

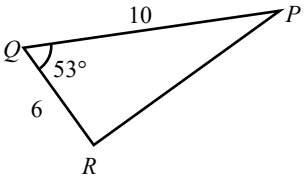
B. $\triangle ABC \sim \triangle XZY$ (*RHS*)

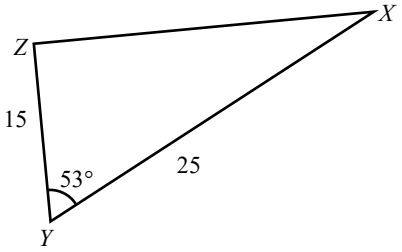
C. $\triangle ABC \sim \triangle XZY$ (*2 sides prop., inc. ∠ equal*)

D. $\triangle ABC \sim \triangle XZY$ (*3 sides proportional*)

30. Which of the following triangles must be similar?







A. $\triangle ABC$ and $\triangle QPR$ only

B. $\triangle PQR$ and $\triangle XYZ$ only

C. $\triangle ABC$ and $\triangle YXZ$ only

D. $\triangle ABC$, $\triangle PQR$ and $\triangle XYZ$

END OF PAPER